3.12 Use the AND operator to compare two binary values

The AND operator is used to compare two binary values, one bit at a time.

The result is 1 only when both bits being compared are 1. Otherwise, the result is 0.

The AND operator is commonly used in computer programming and digital electronics. It can be used to check whether specific bits are set to 1 and to perform operations called bit masking.

Rules for the AND Operator

First Bit	Second Bit	Result
0	0	0
0	1	0
1	0	0
1	1	1

Steps for Using the AND Operator

- Write the two binary numbers one above the other, aligning the digits by place value. Add leading zeros if necessary.
- Compare the bits in each column.
- Write 1 if both bits are 1. Otherwise, write 0.
- Continue until all columns have been compared.

Note: The AND operator does not add or subtract binary numbers. There is no carrying or borrowing.

Examples

Use the AND operator to compare 1101_2 and 1011_2.

Use the AND operator to compare 10110_2 and 01101_2